

SUMMARY:

Applied Mathematics PhD candidate specialized in PDEs, numerical simulation, and high-dimensional modelling. Experienced in Python/C++ scientific computing, optimization, and large-scale numerical experiments on HPC clusters. Interested in quantitative research, AI, scientific computing, and mathematically intensive engineering roles.

TECHNICAL SKILLS

Programming:	Scientific Computing:	Machine Learning:	Mathematics:
• Python (NumPy, SciPy, PyTorch, OpenCV); • C++; • MATLAB; • LaTeX.	• Numerical PDEs; • Optimization; • Monte Carlo; • High-dimensional modelling.	• PyTorch; • Optimization methods; • Data analysis; • HPC clusters.	• Probability; • Statistics; • Linear Algebra; • Numerical Analysis.

EXPERIENCE

- 2022 - 2026; **Researcher in Applied Mathematics** | PhD candidate. [Link to the thesis.](#)
Supervisor: Emeric Bouin *Laboratory:* CEREMADE, Université Paris-Dauphine PSL.
- 2022 - 2026; **Teaching Assistant.** *Establishment:* MIDO, Université Paris-Dauphine PSL.
 - Analysis 1, Bachelor's Degree in Applies Mathematics, 1st year.
 - Analysis 2, Bachelor's Degree in Applies Mathematics, 1st year.
 - Game Theory, Bachelor's Degree in Applies Mathematics, 3rd year.

PROJECTS

- Numerical simulation of the *kinetic Fokker-Planck equation* for gas and plasma. [Link to GitHub](#). Python.
- Numerical simulation of the *run and tumble equation* for chemotaxis. [Link to GitHub](#). Python.
- Jaguar: Energy-efficient navigation of a terrestrial robot in an uncertain environment. [Link to GitHub](#). Python (JAX).
- Web application to play my collection of crosswords and puzzles. [Link to the page](#). HTML, CSS, JavaScript.

SCIENTIFIC PUBLICATIONS

- [1] [L² Hypocoercivity methods for kinetic Fokker-Planck equations with factorised Gibbs states.](#)
- [2] [Sub-exponential tails in biased run and tumble equations with unbounded velocities.](#)
- [3] [Convergence to a non-explicit steady state in non-factorized kinetic Fokker-Planck equations with \(very\) weak velocity confinements.](#)
- *Developed analytical proofs and implemented large-scale numerical PDE simulations on HPC infrastructure.*

EDUCATION

- 2022 - 2026; **PhD, Mathematics**, Université Paris-Dauphine PSL (Paris).
Supervisor: Emeric Bouin *Laboratory:* CEREMADE
- 2020 - 2022; **Master degree, Mathematics**, MAPPA double degree program operated by University of Padova (Italy) and University Paris Sciences & Lettres (France)
Title: Kinetic Theory and Hypocoercivity. *Final grade:* 110/110 cum laude.
- 2017 - 2020; **Bachelor degree, Mathematics**, Università di Padova, Italy.
Title: Rademacher and Whitney's Theorems *Final grade:* 110/110 cum laude
- 2013 - 2017; **High school degree**, Liceo scientifico Copernico-Pasoli, Verona, Italy.

LANGUAGES English (fluent), French (fluent), Italian (native)

INTERESTS Piano, Athletics, Enigmatics.